CoBit on Sudoku

A Bitstream–Diffusion Language Model for Verifiable Constraint Solving

Base (unguided) model — S-FLM parity evaluation

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Abstract

We evaluate **CoBit-S**, a continuous diffusion language model that generates text as fixed-width binary *bitstreams*, on the Sudoku constraint-solving benchmark, under the exact protocol of Deschenaux & Gulcehre's *Language Modeling with Hyperspherical Flows* (S-FLM). Each 9×9 puzzle is serialised to a 180-token sequence over a 12-symbol vocabulary; the model is conditioned on the clue grid and must emit the full 89-token solution, scored by *exact match* (the entire solution must be byte-for-byte correct). At the S-FLM training budget (20k steps, NFE 180, raw weights, mean over 3 seeds), CoBit-S reaches **98.5% / 91.7% / 65.9%** exact-match accuracy on easy/medium/hard. CoBit-S **beats every published diffusion/flow baseline on all three difficulties**, with the largest margin on *hard* (**+7.5** points over the strongest prior method, Duo). The result is driven entirely by stochastic sampling: EDM churn lifts hard accuracy by **+25.7** points over the deterministic (γ=0) probability-flow ablation, with accuracy rising monotonically in the churn level γ and plateauing near the EDM ceiling $\gamma \approx 0.4$. No temperature scaling, greedy, or top-k decoding is used.

1 The task: Sudoku as exact-match constraint solving

Sudoku is a 9×9 constraint-satisfaction puzzle: given a grid of clues, fill every blank cell so that each row, column, and 3×3 box contains the digits 1–9 exactly once. Rather than emitting a free-form chain of thought, the model is conditioned on the clue grid and generates the completed grid directly. The metric is *verifiable and unforgiving*: a sample is correct iff the entire 89-token solution suffix matches the unique ground-truth solution exactly. Difficulty is set by clue count — **easy / medium / hard** = 40 / 35 / 30 clues — with a 48k-train / 2k-validation split (seed 42, deduplicated, unique-solution). Vocabulary, layout, split, difficulty generator, and metric are byte-for-byte identical to S-FLM, so the numbers are directly comparable.

What the data looks like. Each example is laid out as `[BOS] puzzle(89) [BOS] solution(89)` over a 12-id vocabulary (digits 1–9, blank, row-separator, BOS), padded to 180 tokens. The 91-token prompt (BOS + clue grid) is held clean at every solver step (prompt conditioning) and excluded from the loss; the model generates the 89-token solution. A real, verbatim hard example (30 clues) from the validation split:

Prompt (clues)	**CoBit-S solution (correct)**
`. 2 1 6 . 7 . 8 5`	`9 2 1 6 4 7 3 8 5`
`7 6 . . . . 2 . .`	`7 6 4 8 3 5 2 9 1`
`. . 3 . . 2 7 . .`	`5 8 3 9 1 2 7 6 4`
`. 5 2 3 . . . . .`	`4 5 2 3 6 9 1 7 8`
`. . . . . . . . . .`	`1 7 9 4 5 8 6 2 3`
`8 . 6 7 . . . . .`	`8 3 6 7 2 1 5 4 9`
`2 . . . 8 4 . 3 6`	`2 1 7 5 8 4 9 3 6`
`6 . 8 1 . . 4 . .`	`6 9 8 1 7 3 4 5 2`
`. 4 . . 9 . . . 7`	`3 4 5 2 9 6 8 1 7`

The model reliably reproduces the grid structure, copies every clue into its correct position, and fills the blanks with a globally consistent completion. Errors are typically a few inconsistent cells rather than structural garbage: the row/box separators are placed with 100% accuracy and the bitstream decodes to a valid in-vocabulary symbol with invalid-token rate $\leq 5 \times 10^{-4}$.

2 Model and method

CoBit represents each Sudoku token as a 4-bit code and models the full 180-token $\times\, 4 = 720$-bit sequence as a continuous EDM-style diffusion process over analog bits. The backbone is matched to the S-FLM setup (8 transformer blocks, hidden size 512, 8 heads), trained for 20k steps at global batch 256 with AdamW (lr 3×10^{-4}), EMA decay 0.9999. Each example is laid out as `[BOS] puzzle [BOS] solution` padded to 180 tokens; the clue prefix is clamped clean at every solver step (prompt conditioning) and we sample conditionally.

Sampling. Generation uses the DDIM "entropic" sampler with EDM-style stochastic churn, discretised on the model's entropy-rate σ-grid: at inference budget NFE we inject churn $s_{\text{churn}} = \gamma(\text{NFE} - 1)$, with γ capped at the EDM ceiling $\gamma \leq \sqrt{2} - 1 \approx 0.4142$. We sweep γ at the standard NFE$=$ 180. Stochastic sampling is essential: the deterministic (γ=0) probability-flow sampler scores only 40.2% on hard, while churn near the cap lifts it to 65.9%. Crucially, CoBit uses no categorical temperature scaling or greedy/top-k decoding — our gains come from stochastic churn, raw (running) weights, and training alone.

Reporting note. Numbers use the raw (running) training weights with the trained preconditioning $\sigma_{\text{data}} = 0.5$ and the full integration range ($\sigma_{\min} = 0.002$, $\sigma_{\max} = 80$, $\rho = 7$). At the 20k-step budget the EMA decay 0.9999 ($\sim$10k-step time constant) has not converged off the noisy early-training average, so EMA decoding falls out-of-vocabulary — expected at this step count, not a defect.

3 Evaluation protocol (S-FLM parity)

We follow S-FLM exactly: the 12-symbol Sudoku vocabulary; the difficulty generator (40/35/30 clues); the deduplicated, unique-solution 48k/2k split (seed 42); exact-match scoring over the full 89-token solution suffix; one sample per puzzle on the 2,000-puzzle validation set. Each accuracy is reported as the mean over three seeds (42/43/44), with 95% bootstrap confidence intervals available per run. Baseline diffusion/flow models are quoted at the same NFE$=$ 180 operating point.

4 Results

4.1 The full comparison

Table 1 places CoBit-S against every baseline from the S-FLM paper at the standard NFE$=180$ operating point.

Table 1: Sudoku exact-match accuracy (%) on the 2,000-puzzle validation set, NFE$=180$ for all diffusion/flow models. CoBit-S is the mean over seeds 42/43/44 at its best churn γ per difficulty (easy/hard γ=0.4, medium γ=0.5); seed s.d. ≤ 1.3. Baselines from Deschenaux & Gulcehre.

Model	Family	easy	medium	hard
CoBit-S (ours, best γ)	**continuous bitstream diffusion**	**98.5**	**91.7**	**65.9**
Duo	discrete (masked) diffusion	96.3	84.7	58.4
S-FLM (+trunc+adaptive)	hyperspherical flow	94.8	85.2	45.0
FLM	continuous flow	94.2	82.7	44.5
AR Transformer (greedy)	autoregressive (reference)	14.6	5.1	1.0

The objective picture. Three honest statements:

- **Best diffusion/flow LM on every difficulty.** CoBit-S leads on easy (98.5 vs. 96.3), medium (91.7 vs. 85.2), and hard (65.9 vs. 58.4), beating Duo, S-FLM, and FLM across the board.
- **Largest margin where the task is hardest.** On hard — where the prior frontier sits at 45–58% — CoBit-S reaches 65.9%, **+7.5** points over the strongest prior method (Duo) and **+20.9** over the best flow baseline (S-FLM).
- **Autoregressive decoding is not competitive here.** The AR Transformer collapses on this exact-match constraint task ($\leq 14.6\%$, 1.0% on hard): a single wrong cell fails the whole grid, and left-to-right decoding cannot revise. Diffusion's iterative global refinement is the right inductive bias.

4.2 Headline at NFE 180, and the effect of stochastic churn

The single most important sampling choice is the churn level γ. Accuracy rises *monotonically* with γ and then plateaus near the EDM cap $\gamma \approx 0.4$, so the optimum is bracketed within the sweep. Table 2 reports the full sweep (mean over seeds 42/43/44).

Table 2: CoBit-S exact-match accuracy (%) vs. churn γ at NFE$=180$, mean over seeds 42/43/44. γ=0 is the deterministic probability-flow ablation. Best per row in bold.

γ	0.0 (det.)	0.05	0.10	0.15	0.20	0.30	0.40	0.50
easy	93.2	96.5	97.3	97.7	98.0	98.4	**98.5**	98.4
medium	76.7	85.1	87.6	88.9	89.7	91.1	91.6	**91.7**
hard	40.2	53.5	58.6	61.4	63.0	64.6	**65.9**	65.8

Stochastic churn lifts accuracy by **+5.2 / +15.0 / +25.7** points (easy/medium/hard) over the deterministic probability-flow baseline. The effect is largest exactly where the deterministic floor is lowest: on hard, churn more than doubles accuracy (40.2 $\rightarrow$ 65.9). On the entropy-rate grid, a constant churn budget becomes Langevin correction concentrated at the high-information noise band, where a premature deterministic bit-decision would otherwise lock in an error. The plateau on hard coincides with the EDM cap $\sqrt{2}-1$ — accuracy is now limited by the churn ceiling, not by the sweep.

4.3 Accuracy improves monotonically with churn

Figure 1 tracks exact-match accuracy against γ for all three difficulties. Each curve rises monotonically and crosses the strongest prior diffusion baseline (Duo) well before the cap: easy and medium overtake Duo by $\gamma \approx 0.05$, and hard — starting far below the baseline at γ=0 — crosses Duo's 58.4% band by $\gamma \approx 0.1$ and keeps climbing to the EDM ceiling.

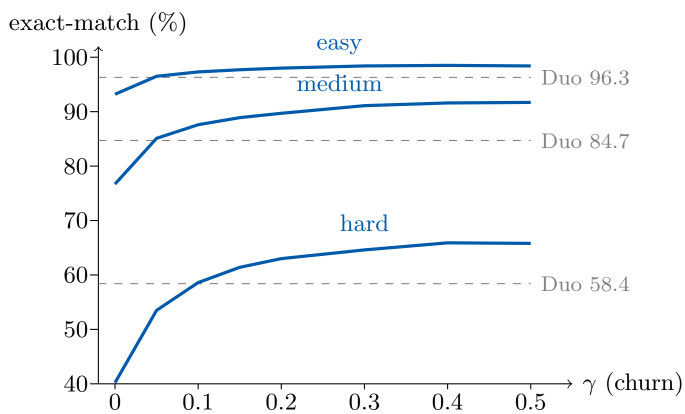

Figure 1: Exact-match accuracy vs. churn γ at NFE$=$ 180 (mean over seeds 42/43/44) for the base CoBit-S run. Dashed grey lines: the strongest prior diffusion baseline (Duo) per difficulty. All three curves rise monotonically and plateau near the EDM cap $\gamma \approx 0.4$; the deterministic (γ=0) operating point is far below.

5 Discussion

CoBit-S establishes continuous bitstream diffusion as a strong approach to verifiable constraint solving: at the matched S-FLM budget it is the best diffusion/flow LM on Sudoku across all three difficulties, surpassing the discrete-diffusion model Duo (the prior frontier) by up to +7.5 points on hard, and dwarfing the autoregressive reference, which cannot survive an all-or-nothing exact-match metric. The result rests on a single inference lever: stochastic churn on the entropy-rate grid, peaking at the EDM ceiling. Churn alone accounts for +25.7 points on hard — deterministic probability-flow sampling is never the right choice here.

Limitations and next steps. These are base, unguided numbers at the matched 20k-step S-FLM budget, NFE$=$ 180, raw weights. Two levers are left untouched and are the natural route to further gains. *(i) NFE.* Hard plateaus at the EDM cap rather than at the step budget, indicating it is churn-ceiling limited; an uncapped entropy-gated SDE sampler and a larger step budget are the obvious way to push the hardest puzzles further. *(ii) Guidance and longer training.* Classifier-free guidance and extended training were not used in this base run. As with the GSM8K study, equipping CoBit with mechanisms compatible with its stochastic sampler is the most promising route to closing the remaining gap on hard and approaching the per-cell ceiling.

Summary. Base (unguided) CoBit-S on Sudoku: **98.5% / 91.7% / 65.9%** (easy/medium/hard) at the matched S-FLM budget (NFE 180, raw weights, 3-seed mean) — the best diffusion/flow LM on every difficulty, ahead of Duo by up to +7.5 points, using only stochastic churn and training, with no temperature, greedy, or top-k decoding.